

Research Article

An Early Miocene regurgitalite containing basking shark remains from Slovenia: insight into cetorhinid paleoecology of the Central Paratethys

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Abstract

A bromalitic concentration from the Lower Miocene (upper Burdigalian, calcareous nannoplankton Zone NN4) deposits of Meljski Hrib near Maribor, northeastern Slovenia, preserves at least 25 fragmentary basking shark gill rakers together with polygonal calcified-cartilage tesserae and disarticulated echinoid remains. Comparative morphology, geometric morphometric analysis of gill-raker bases, and micro-computed tomography of associated skeletal elements support assignment of the material to *Keasius* cf. *K. septemtrionalis*. This represents the first confirmed occurrence of *Keasius* from Slovenia and extends the known paleogeographic distribution of the genus into the southeastern margin of the Central Paratethys. The close association of cetorhinid remains with abundant echinoid material and the absence of acid etching on gill-raker surfaces indicate that the assemblage is rather a regurgitalite than a coprolite. The associated invertebrate fauna indicates deposition in an outer-shelf to upper-slope environment (approximately 150–250 m water depth). This taphonomic and paleoenvironmental context is most consistent with opportunistic scavenging by a benthopelagic vertebrate predator, plausibly a lamniform or hexanchid shark. The assemblage provides rare fossil evidence of trophic interactions involving planktivorous basking sharks and macrophagous vertebrates in Miocene deep-water ecosystems and provides new insights into the paleoecology and biogeographic history of cetorhinid sharks within the Paratethyan realm.

Keywords

Bromalite, Central Paratethys, cetorhinid, geometric morphometrics, Miocene, palaeoecology, scavenging, trophic interactions

Introduction

Recent basking sharks (family Cetorhinidae) are large plankton-feeding lamniform sharks that filter huge volumes of water through their gills (Compagno et al. 2005). Cetorhinids are represented in the fossil record primarily by isolated teeth, gill rakers, dermal scales, and vertebrae, with partial skeletons being rare (Cappetta et al. 2016; Hovestadt and Hovestadt-Euler 2012; Reinecke et al. 2015; Welton 2013a). The family includes three genera: the extant *Cetorhinus* Blainville, 1816, the extinct *Keasius* Welton, 2013, and *Caucasochasma* Prokofiev & Sychevskaya, 2018. Together, these taxa document the evolutionary history of planktivory among lamniform sharks (Kovalchuk and Barkaszi 2021).

The genus *Cetorhinus* first appears in the fossil record during the Early to Middle Miocene (Malyshkina et al. 2022; Welton 2013b) and is represented today by a single species, *Cetorhinus maximus* (Gunnerus, 1765), which inhabits temperate and boreal waters worldwide (Hoelzel et al. 2006; Rigby et al. 2021). Two extinct species of the genus are recognized: *Cetorhinus huddlestoni* Welton, 2014 from the Middle Miocene of California (USA) and South Korea (Malyshkina et al. 2022), based on isolated teeth, and *C. piersoni* Welton, 2015 from the Lower Miocene of Oregon (USA), also based on teeth alone.

The extinct genus *Keasius* ranges from the Middle Eocene to Lower Miocene; it was erected by Welton (2013a) to accommodate a fossil basking shark formerly assigned to *Cetorhinus*. Four species of *Keasius* are currently recognized: *K. taylori* Welton, 2013 from the Upper Eocene of Oregon (USA), *K. parvus* (Leriche, 1908) from the Lower Oligocene of Belgium, *K. septemtrionalis* Reinecke, von der Hocht & Dufraing, 2015 from the Upper Oligocene–Lower Miocene of Germany, and *K. rhenanus* Reinecke, von der Hocht & Dufraing, 2015 from the Lower to Middle Miocene of Germany. Fossil remains of *Keasius* are widespread, occurring in Europe, North America, Antarctica, and Asia (Bienkowska-Wasiluk 2021; Cappetta et al. 2016; Cione and Reguero 1998; Kovalchuk and Barkaszi 2021; Malyshkina 2006; Nazarkin 2014; Reinecke et al. 2015; Welton 2013a). An articulated partial skeleton from the Lower Oligocene of Germany (Hovestadt and Hovestadt-Euler 2012) provided associated skeletal material that confirmed the anatomical connection between dentition and gill rakers. However, this was already demonstrated by the holotype of *K. taylori*, which comprises teeth, vertebrae, and gill rakers from a single individual.

Caucasochasma is monospecific, represented by *C. zherikhini* from the Lower Oligocene of the North Caucasus, Russia (Prokofiev and Sychevskaya 2018). Prokofiev and Sychevskaya (2018) interpreted it as a bottom-associated planktivorous shark that fed on plankton near the sea floor, representing a unique ecological adaptation within the family.

Thus, cetorhinids have a nearly continuous fossil record from the Middle Eocene to the present, documenting the gradual re-establishment of pelagic suspension feeding among lamniform sharks. Importantly, cetorhinids were not the first lamniforms to evolve suspension feeding; Cretaceous taxa such as *Pseudomegachasma* and *Aquilolamna* also show adaptations indicating planktivory (Shimada et al. 2015; Vullo et al. 2021). The Oligocene diversification of *Keasius* parallels the radiation of baleen-assisted mysticetes (Fitzgerald 2006; Tsai et al. 2024), reflecting broader restructuring of plankton-based marine ecosystems during a period of increasing oceanic productivity (Bartol et al. 2014).

Here we describe basking shark remains and associated faunal elements retrieved from a bromalitic concentration from the Lower Miocene of northeastern Slovenia and analyze these findings in the context of trophic interactions in a deep-water Paratethyan environment.

Geological setting

The bromalite specimen described herein was recovered from yellowish marls forming a talus deposit in Meljski Hrib (46°33.63'N, 15°40.55'E) near Maribor, north-eastern Slovenia (Fig. 1). The locality is well known for Miocene invertebrates and fossilized wood (Mikuž and Gašparič 2014). Fossiliferous beds form steep cliffs up to 100 m high. The lower part of the succession consists of dark grey siltstones and sandstones with scarce macrofossils (Gašparič and Halasova 2015; Gašparič et al. 2025), while the overlying indurated light grey to yellowish marls contain calcite-rich concretions and yield a sparse deep-water macrofauna of bivalves, echinoids, and mass mortality assemblages of brittle stars (Gašparič et al. 2021).

These sediments belong to the Pannonian Basin System, specifically the Slovenske gorice tectonic unit. They were deposited within the Mura Depression, part of the western Mura–Zala Basin of the Central Paratethys (Fodor et al. 2002). During the Early Miocene, synrift subsidence produced deep marine basins within the Paratethys region (Pavšič and Horvat 2009; Rifelj and Jelen 2001). Marine strata of the Central Paratethys in Slovenia range in age from the Early Miocene (Burdigalian/Karpatian) to the Middle Miocene (Serravallian/Sarmatian) (Pavšič and Horvat 2009; Žnidarčič and Mioč 1989). The stratigraphy is constrained by foraminiferal and calcareous nannoplankton assemblages (Bartol 2009; Bartol et al. 2013; Gašparič and Halasova 2015; Rijavec 1976, 1978). Nannoplankton analysis of the matrix places the sediments within the lower part of Zone NN4, i.e., the upper Burdigalian (Ottangian–Karpatian regional stage). This stratigraphic attribution provides a clear temporal framework for the bromalite and supports its Early Miocene age.

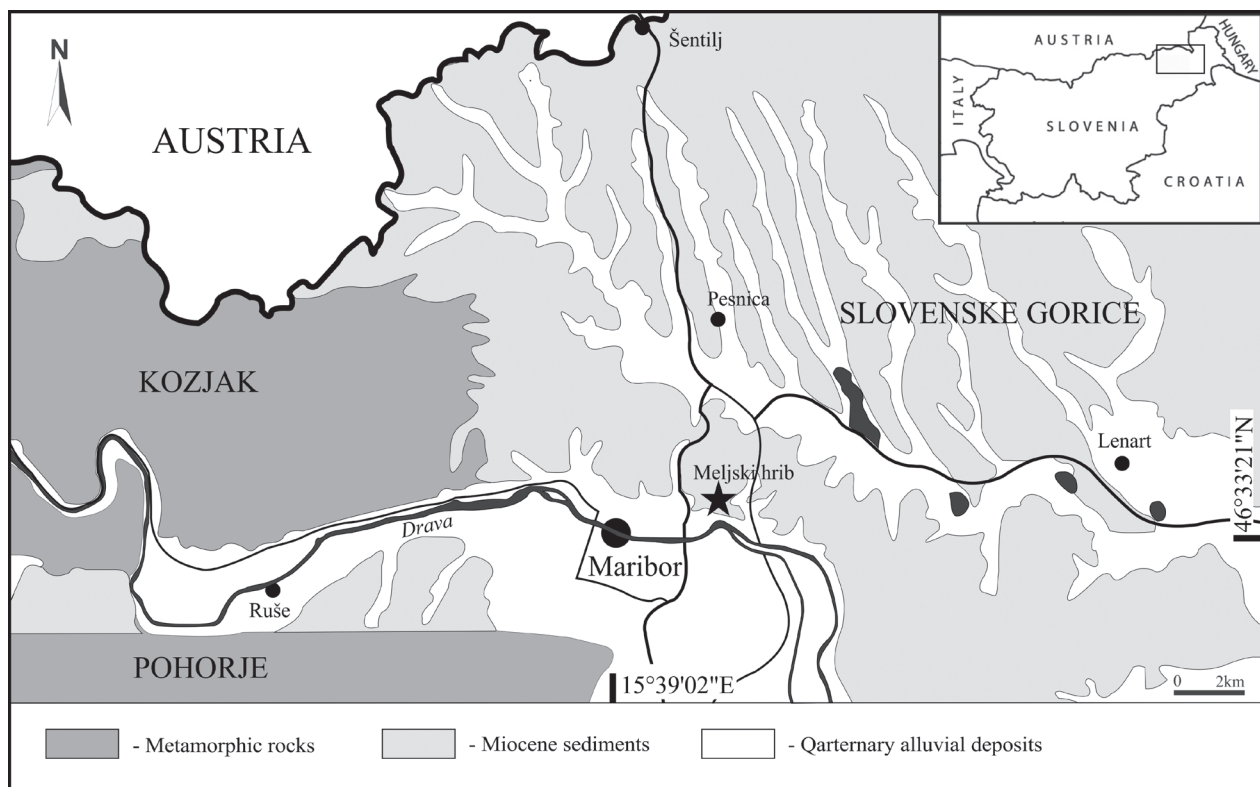


Figure 1. A simplified geological map of northeastern Slovenia. The locality of Meljski Hrib is marked with an asterisk.

Materials and methods

The studied specimen represents a bromalitic concentration that includes numerous fragmented gill rakers associated with abundant remains of irregular spatangoid echinoids. The echinoid material consists of disarticulated test fragments and hundreds of complete and partial spines, indicating the breakdown of several individuals. The concentration also includes foraminifera, among which *Strictocostella* Patterson, 1987 was recognized, as well as small polygonal plates with tessellated structure. Because the echinoid material occurs within the bromalite itself rather than in the surrounding sediment, its inclusion has implications for identifying the producer and supports exclusion of certain predators (e.g., crocodiles), whose digestive processes would have dissolved echinoid tests before regurgitation.

Fossil elements were identified using diagnostic morphological features and comparative analyses of both fossil and extant representatives of the respective taxa. Preparation was carried out under Leica SZ4 and EZ4D stereomicroscopes. Photographs were taken with a Nikon Z7II digital camera fitted with a Nikkor Z MC 50 mm f/2.8 lens. Image stacks were processed using Helicon Focus and subsequently refined in Adobe Photoshop CS6. Micro-computed tomography (μ -CT) imaging was performed with a Phoenix L240 system developed by GE Sensing & Inspection Technologies at

the Earth Science Institute of the Slovak Academy of Sciences in Banská Bystrica. We follow the descriptive terminology of gill rakers after Welton (2013a, fig. 11) and comparative morphological characters compiled by Kovalchuk and Barkaszi (2021, Table 1). Additionally, three gill rakers with discernible outlines (SLO, $n = 3$) were included in a 2D landmark analysis of gill-raker shape variation following Kovalchuk and Barkaszi (2021) and compared against samples of *Cetorhinus maximus* ($n = 14$, recent), *Keasius taylori* ($n = 5$, Eocene), *K. parvus* ($n = 33$, Lower Oligocene–Upper Miocene), *K. rhenanus* ($n = 7$, Lower–Upper Miocene), and *K. septemtrionalis* ($n = 10$, Upper Oligocene). The *Keasius* samples contain the respective type specimens (*K. taylori* – 4 holotypes, *K. parvus* – 4 syntypes) or those associated with type series (*K. rhenanus*, *K. septemtrionalis*). Landmark data were acquired using tpsUtil and tpsDig, and analyzed (Procrustes fit, principal component analysis, and canonical variates analysis) in MorphoJ (Klingenberg 2011). One-way PERMANOVA of PCA scores was carried out in PAST 4.17 (Hammer et al. 2001).

Institutional abbreviations

RGA/SMNH Paleontological collection of the Institute for Palaeobiology and Evolution, Slovenia.

Systematic Paleontology

Class Chondrichthyes Huxley, 1880

Subclass Elasmobranchii Bonaparte, 1838

Superorder Galeomorphii Compagno, 1973

Order Lamniformes Berg, 1958

Family Cetorhinidae Gill, 1862

Genus *Keasius* Welton, 2013

***Keasius* cf. *K. septemtrionalis* Reinecke, von der Hocht & Dufraing, 2015**

Figs 2–4

Referred material. RGA/SMNH 2736, a bromalitic concentration of fragmented marine fauna including at least 25 basking shark gill rakers, two nearly complete, four basal fragments, and numerous filament sections.

Description. Twenty-five fragmentary gill rakers are preserved, six retaining basal portions. The longest incomplete raker measures more than 35 mm, although it has the shortest base and the thinnest filament among the preserved elements. The filaments are gently curved medially, compressed laterally, and oval in cross section. They are flat to weakly concave at their widest point, where the filament joins the base. Both oral and aboral margins are smoothly rounded. A median longitudinal trough is present, weak near the base connection and more pronounced apically. The surface is smooth to weakly wrinkled near the base and becomes densely, finely striated toward the apex.

The base of the gill rakers bears a well-developed, acute, triangular distal projection, a straight basal edge, and a rounded basal angle. The medial process is long and sickle-shaped in all specimens, tapering apically and exhibiting a low basal height. Its mesial edge is

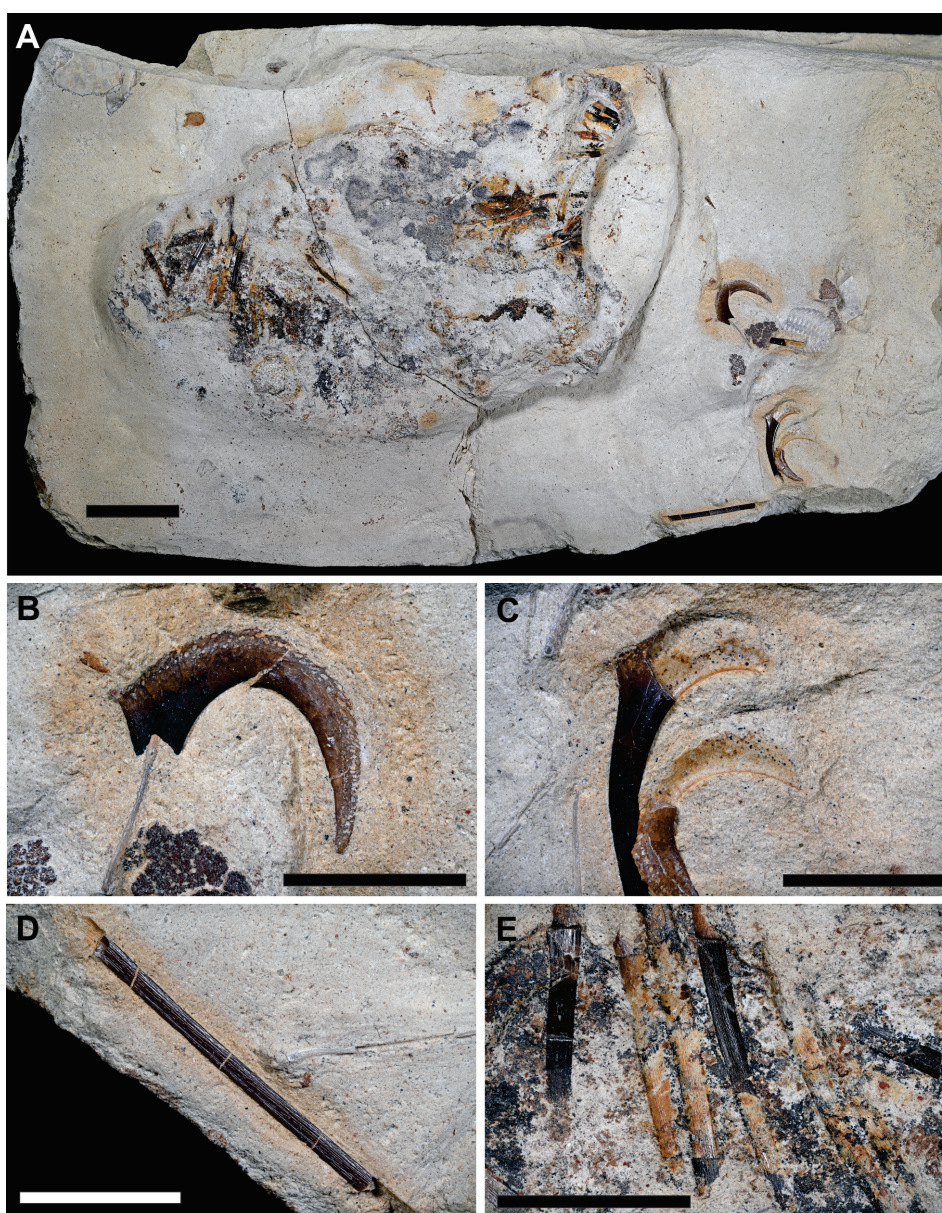


Figure 2. A. Bromalitic concentration of fragmented *Keasius* cf. *K. septemtrionalis* gill rakers; B. Detail of base; C. Base/filament attachment; D. Close-up of striated filament; E. Fragments of seven different filaments. Scale bar: 10 mm (A); 5 mm (B–E).

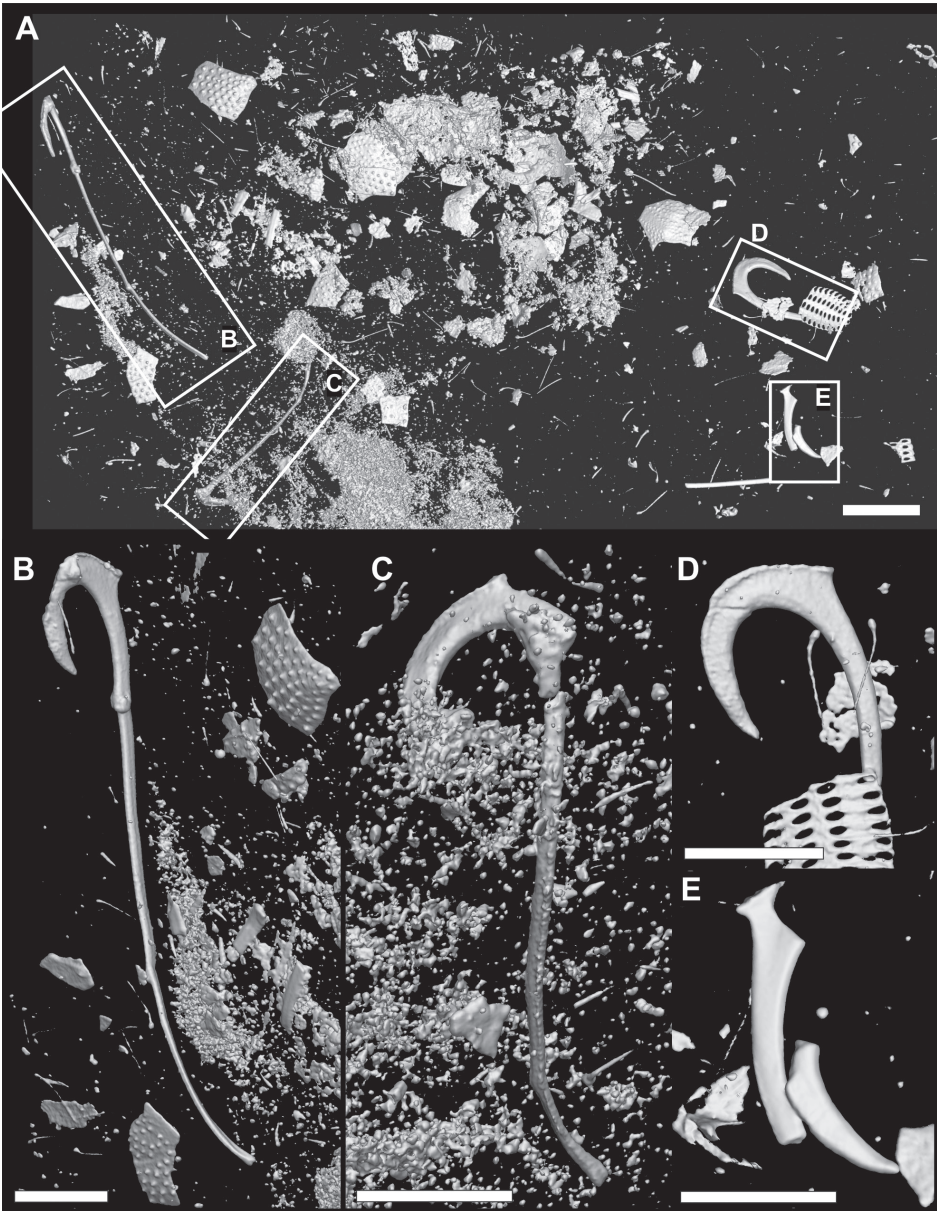


Figure 3. Micro-computed tomography image of the bromalitic specimen with *Keasius* cf. *K. septemtrionalis* gill rakers. **A.** Complete gill raker; **B.** Complete gill raker; **C** detail of base; **D.** Broken base/filament attachment. Scale bar: 10 mm (**A**); 5 mm (**B–E**).

Table 1. Procrustes (D_p) and Mahalanobis distances (D_M) between the analyzed samples ($p < 0.05$, pair-wise one-way PERMANOVA).

D_p / D_M	<i>Cetorhinus maximus</i>	<i>Keasius parvus</i>	<i>Keasius rhenanus</i>	<i>Keasius septemtrionalis</i>	<i>Keasius taylora</i>	SLO material (herein)
<i>Cetorhinus maximus</i>	–	3.8595	4.8497	5.7300	7.0407	9.9453
<i>Keasius parvus</i>	0.1210	–	4.7012	6.3341	5.2910	10.4471
<i>Keasius rhenanus</i>	0.1820	0.2013	–	5.9085	5.0589	10.6934
<i>Keasius septemtrionalis</i>	0.2293	0.2917	0.1449	–	6.9661	9.7350
<i>Keasius taylora</i>	0.2621	0.2156	0.1807	0.2890	–	12.6247
SLO material (herein)	0.2757	0.3369	0.2640	0.2213	0.4200	–

smoothly rounded. A narrow, rugose basal attachment surface extends from the distal basal projection to the tip of the medial process and is perforated by numerous evenly spaced foramina. The bight is rounded and relatively broad. Except for the basal attachment surface, all exposed surfaces of gill rakers are covered with

enameloid. No evidence of acid etching was observed, consistent with regurgitation rather than defecation.

Remarks. The general morphology of gill rakers of representatives of the genus *Keasius* is rather species-specific. Welton (2013a, 2013b) concluded that more derived gill rakers are characterized by less robust and longer

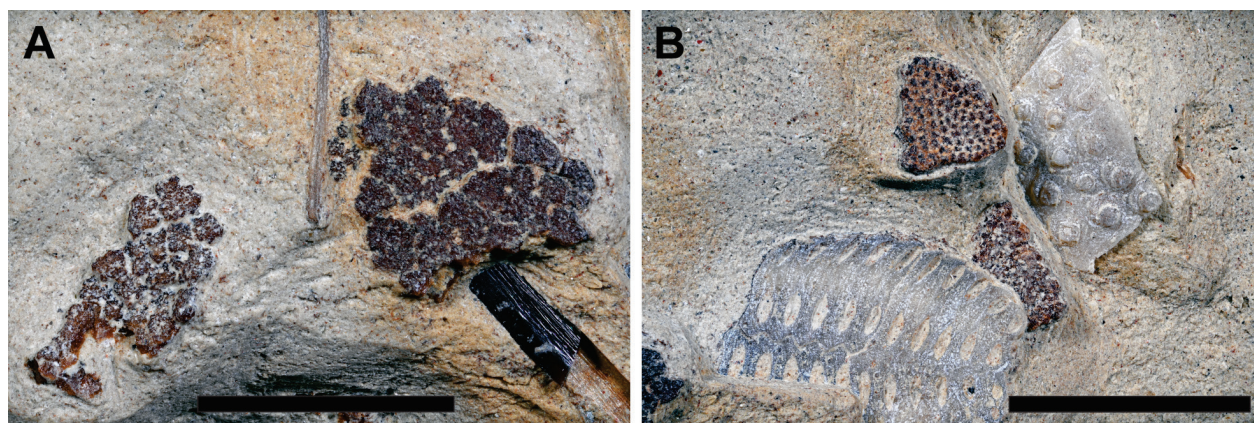


Figure 4. **A.** Calcified cartilage tesserae plates in close proximity to a gill-raker filament fragment and an echinoid spine; **B.** Disarticulated echinoid test fragments together with calcified cartilage tesserae. Scale bars: 5 mm.

bases, longer medial processes, wider bights, and strongly curved filament bases. Kovalchuk and Barkaszi (2021) arrived at the same conclusion based on morphological measurements revealing a clear stratigraphic progression of samples. The gill rakers of the type species, *K. taylori*, which is the earliest species within the genus, have a very short medial process and high basal height, whereas in the younger *K. septemtrionalis* and *K. rhenanus* gill rakers have a longer medial process and lower basal height. The studied gill rakers differ from those of *C. maximus*, which have relatively long medial processes but also larger basal height (Fig. 5).

The low bases and relatively broad bights of the Slovenian specimens indicate that the preserved elements likely originated from the distal part of the gill arch, where similar morphologies are seen in both fossil and modern basking sharks (Reinecke et al. 2015; Welton 2013a). Within *Keasius*, most distinguishing characters are based on dental morphology (Reinecke et al. 2015), whereas gill-raker features vary along the gill arch and show considerable intraspecific overlap. The slender, laterally compressed filaments and the elongate, sickle-shaped medial process of the Slovenian specimens resemble those of *K. septemtrionalis* from the Mainz Basin (Reinecke et al. 2015, fig. 11). The low basal height and broad rounded bight align with the morphometric range recorded for that species and differ from the higher, narrower bases typical of *K. rhenanus* and the narrow bights exhibited by *K. parvus* (Reinecke et al. 2015, fig. 16).

Shape analysis shows that gill rakers of *Keasius rhenanus* and *K. septemtrionalis* are very similar and the two samples greatly overlap in the morphospace of the first two principal components, which describe 72.7% of the total variance (Fig. 6A). The specimens from Slovenia along PC1 are closer to *K. septemtrionalis* (Fig. 6A). The similarity is also evident along CV1 when comparing the different samples (Fig. 6B) and is additionally supported by values of the Procrustes and Mahalanobis distances (Tab. 1). However, the small number of intact specimens

in the studied sample ($n = 3$) and the fact that they most likely come from a single animal prevent an unequivocal species identification.

The biostratigraphic position of the Meljski Hrib deposits, within calcareous nannoplankton Zone NN4 (upper Burdigalian / Ottnangian–Karpatian), provides an additional argument for affinity with *Keasius septemtrionalis*. This species ranges from the Late Oligocene (Chattian) to the Early Miocene (Aquitanian–Burdigalian), whereas *K. parvus* is mostly restricted to the Lower Oligocene and *K. rhenanus* appears slightly later, in the Early to Middle Miocene (Reinecke et al. 2015). The age and geographic position of the Slovenian material thus fall squarely within the distribution of *K. septemtrionalis*, representing the southeasternmost occurrence of this lineage in the Central Paratethys.

Considering all the above, we assume that the preserved gill rakers correspond closely to those in *Keasius septemtrionalis* both in morphology and age, but the small number of intact specimens and the lack of associated dental material hinder a definitive species-level identification. Although the Slovenian specimens, like all *Keasius* taxa, share certain similarities with *Cetorhinus maximus* (e.g., laterally compressed filaments), they differ by many key descriptive diagnostic traits as well as in the morphospace of the geometric shape analysis. Furthermore, the stratigraphic position of the material (upper Burdigalian, NN4) aligns with the known temporal range of *K. septemtrionalis* rather than the Neogene–Recent distribution of *Cetorhinus*. For these reasons, referral to *Cetorhinus* is not supported, and a cautious determination as *Keasius* cf. *K. septemtrionalis* is therefore appropriate.

Associated macrofauna and Paleoecology

The associated fauna at Meljski Hrib indicates an outer shelf to slope environment, with water depths of at least 150 m, but likely exceeding 250 m. Deep-water crustaceans include *Callianopsis jahringensis* (Glaessner,

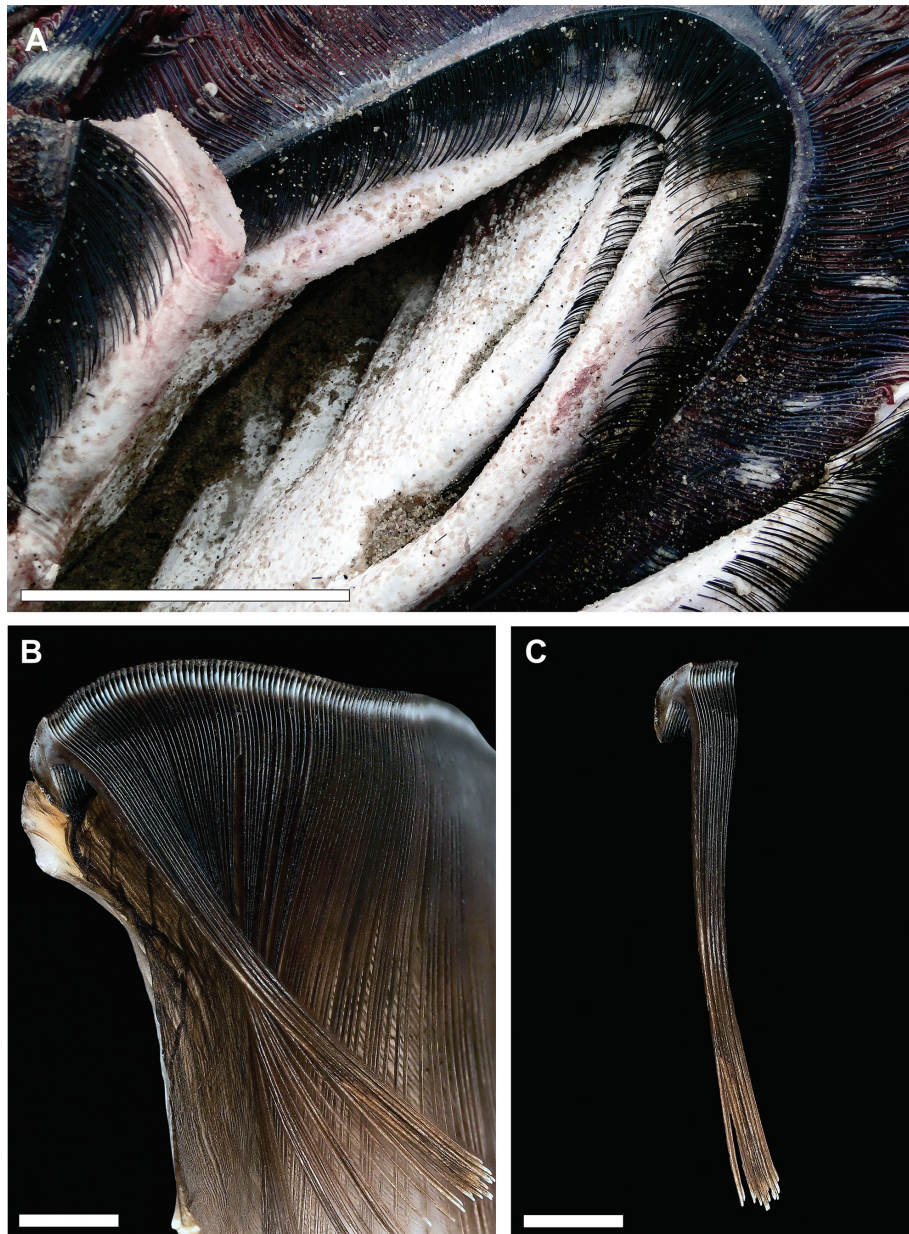


Figure 5. Extant basking shark *Cetorhinus maximus* (Gunnerus, 1765). **A.** Massive cartilaginous gill arches (stranded specimen, Menemsha Beach, Martha's Vineyard Island, Massachusetts, USA); **B.** Each gill arch supports a dense, bristle-like structure made of modified denticles known as gill rakers (Chinese Sea, Pacific Ocean); **C.** Gill raker features a thickened, half-moon base that anchors into a vascularized membrane of the gill arch, transitioning into an elongated, thread-like filament. **B–C:** Spec. no. 7-798/RZ, Shark collection, René Kindlimann, Aathal, Switzerland. Scale bar: 10 cm (**A**); 10 mm (**B**, **C**).

1928), *Metanephrops serendipitus* Gašparič, Tshudy, Chan & Ćorić, 2021, and brachyurans such as *Styrioplax exiguus* (Glaessner, 1928) and *Bathynectes muelleri* Ossó & Stalennuy, 2011 (Gašparič and Halasova 2015; Gašparič and Hyžný 2015; Gašparič et al. 2021, 2025). Mollusks include *Solemya doderleini* Mayer, 1861, *Limaria labani* (Meznerics, 1936), *Lentipecten denudatus* (Reuss, 1867), and *Macoma elliptica* (Brocchi, 1814), all characteristic of deep-water facies (Harzhauser et al. 2011).

The scaphopod *Gadilina taurogracilis* Sacco, 1897, considered bathyal and indicative of unstable deep-marine environments with high sedimentation rates

(Ceregato et al. 2007; Harzhauser et al. 2011), reinforces the deep-water interpretation. The inclusion of echinoid test fragments and spines within the bromalite itself, rather than in surrounding sediment, is significant. Their preservation suggests ingestion together with cetorhinid remains, implying feeding near the seafloor by a benthopelagic predator. This observation strongly disfavors crocodiles, which could not have accessed such depths, as well as pelagic mammals, whose regurgitation at the surface would have dispersed fragile skeletal debris. Instead, the faunal association supports a benthic or demersal predator as the most plausible agent.

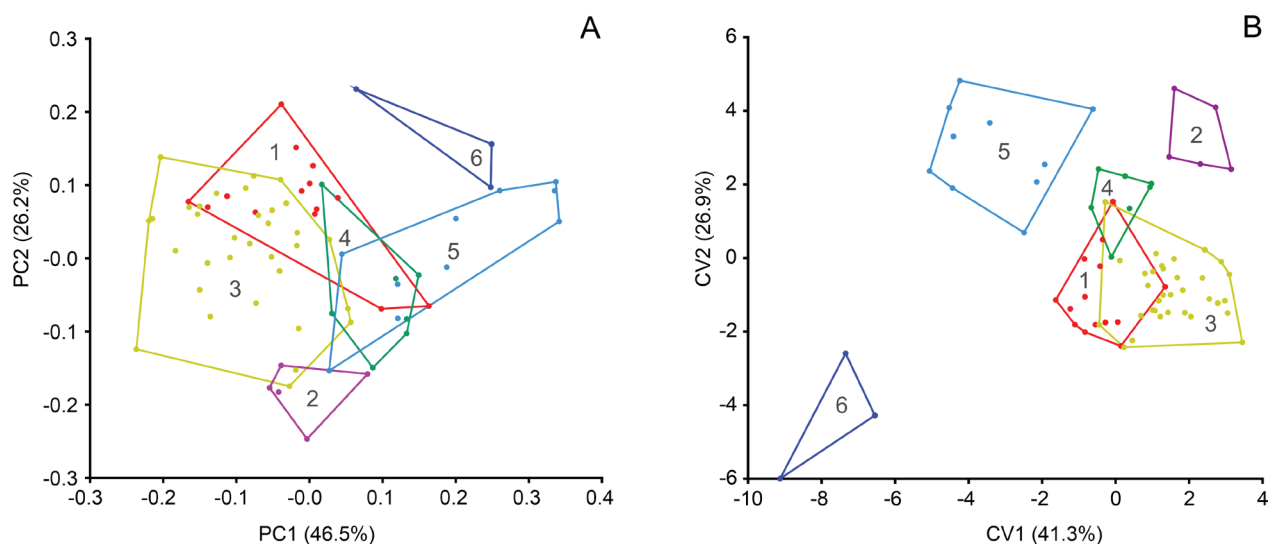


Figure 6. Principal component (A) and canonical variates (B) analyses of the shape of gill rakers of European basking sharks and the studied specimens from Slovenia: 1. *Cetorhinus maximus*; 2. *Keasius taylori*; 3. *Keasius parvus*; 4. *Keasius rhenanus*; 5. *Keasius septemtrionalis*; 6. studied specimens, SLO.

Alternative explanations, including regular shedding of gill rakers (Parker and Boeseman 1954; Van Deinsen and Adriani 1953; Sims 1999, 2008), carcass decomposition, or hydrodynamic concentration, are acknowledged. However, the intimate association of gill rakers, tesseræ, and echinoid material favors the regurgitalite interpretation. While alternative scenarios remain viable, the stratigraphic and ecological context of the deposits supports scavenging by benthopelagic sharks as the most parsimonious explanation.

Discussion and conclusions

The bromalite described herein contains abundant fragmentary basking-shark gill rakers, disarticulated echinoid test fragments and spines, foraminifera, and polygonal calcified-cartilage tesseræ (Fig. 4A, B). The tessellated cartilage plates correspond to the branchial basket of a basking shark (Hovestadt and Hovestadt-Euler 2012; Kemp and Westrin 1979), supporting the interpretation that the material originated from a decaying carcass. Direct predation on basking sharks cannot be entirely excluded (Matthews and Parker 1950), but the fragile preservation of gill rakers and absence of acid etching support ingestion during opportunistic scavenging followed by regurgitation (Walker and Brett 2002).

The co-occurrence of echinoid remains within the bromalite itself is critical, as it demonstrates ingestion near the seafloor. This effectively disfavors crocodiles, which could not have accessed such depths, and pelagic mammals, whose regurgitation at the surface would have dispersed fragile skeletal debris before deposition. Instead, benthopelagic sharks (particularly other Lamniformes or Hexanchidae) represent the most plausible producers. These taxa are known opportunistic feeders (Belben et al. 2017; Reum et al. 2020), potentially capable of ingesting both cetorhinid remains and benthic invertebrates (e.g., echinoids).

Despite the limited number of intact gill-rakers, the preserved material exhibits a combination of morphological and morphometric traits most consistent with *Keasius septemtrionalis*. Stratigraphically, the late Burdigalian (NN4) age of the Meljski Hrib deposits coincides with the known temporal range of this species (Reinecke et al. 2015). We therefore adopt the cautious designation as *Keasius* cf. *K. septemtrionalis*.

In the Early Miocene of the Central Paratethys, a basking shark carcass would have represented as substantial energy-rich resource, attracting scavengers. If the carcass remained buoyant, pelagic scavengers would have dominated (Smith and Baco 2003). Once settled on the seafloor, benthic invertebrates and demersal fishes would have taken over (Higgs et al. 2014), while benthopelagic sharks exploited the remains. Extinct lamniforms such as *Carcharodon hastalis* and *Otodus megalodon* are well documented from the Miocene of Slovenia (Mikuž and Šoster 2013; Mikuž et al. 2013). Though lacking serrated teeth, *Carcharodon hastalis* possessed large, robust, smooth-edged teeth adapted for grasping and slicing soft tissue (Dubikovska et al. 2025), while *Otodus megalodon* was equipped to dismember large prey or carcasses (Shimada et al. 2025, 2026). Hexanchid sharks, as demersal taxa, also represent a viable candidate, since they could have ingested cetorhinid remains together with echinoids in deep-water settings (Ebert 2002; Reum et al. 2020).

While direct fossil evidence of scavenging on basking sharks is lacking, documented cases of lamniform teeth embedded in marine mammal bones indicate opportunistic feeding or scavenging behavior (Godfrey et al. 2025). The bromalitic nature of the specimen thus records a rare instance of trophic interaction among vertebrates in marine environments. The combination of cetorhinid raker fragments, tessellated cartilage plates, and deep-water echinoids points to deposition within an outer-shelf to upper-slope setting (Cappetta et al. 2016; Gašparič

et al. 2025). This assemblage provides rare indirect evidence of the presence of basking sharks (*Keasius* cf. *K. septemtrionalis*) in the Central Paratethys and indicates that benthopelagic sharks, including lamniforms and hexanchids, could have exploited their carcasses. The described specimen illustrates the ecological complexity of Miocene marine food webs and highlights the role of scavenging within deep-water ecosystems.

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Additional information

Conflict of interest

The authors have declared that no competing interests exist.

Ethical statement

No ethical statement was reported.

Artificial Intelligence (AI) use

The authors accept full responsibility for the content of the manuscript, including the disclosure of any use of AI.

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Author contributions

RG, OK, MM and MK devised the project, the main conceptual ideas and the proof outline. TH, MKV, TK and ZB contributed to the interpretation of the results. MM prepared the microtomography images, RG handled the macrophotography and prepared the specimen figures. ZB conducted the morphometric and landmark analysis and the interpretation of the results. RG took the lead in writing the manuscript. All authors provided critical feedback, contributed to revisions and approved the final version.

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Data availability

The data supporting the findings of this study are available within the article. Micro-computed tomography (microCT) datasets are archived in Zenodo and will be made publicly available upon acceptance.